Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/04/2008
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	The Melting of Aqueous Foams		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
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Task Description:

The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology.

There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into ASL technology for NASA's strategic exploration mission.

The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability.

The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ASL, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations.

During the last academic year my group consisted of Mr. Alex Gittings and Dr. Klebert Feitosa. Together we published one peer-reviewed paper, we submitted one which is now under review, and we are writing a third. I gave one invited conference talk and one seminar; Klebert gave two contributed conference talks and several job talks. And we made significant progress on several different aspects of the overall FOAM project:
(1) Reviews. I attended the Delta Phase B meeting held at ESTEC headquarters in Noordwijk (Netherlands), and I contributed to the collocation meeting held in Bordeaux (France) to resolve apparent exceptions of the engineering plans to meet the scientific requirements.

(2) Steady-State Foams. The behavior of wet foams is notoriously difficult to study on earth due to rapid gravitational drainage. This problem is exacerbated by coarsening, the diffusion of gas from smaller to larger bubbles. In order to push our understanding of these effects, and to provide a baseline estimate for behaviors to be more thoroughly studied in microgravity, we created steady-state foams in which gas is continuously bubbled into a pool of surfactant solution. Then we measured the liquid fraction, using conductivity, and the bubble size distribution, using digital photographs and image analysis of surface bubbles. Since the system is in steady state, this is done at out leisure as a function of height in the column of foam. Prior evolution studies all are done as a function of time and are necessarily more rushed and irreproducible. With our apparatus we acquired four complete data sets at four different gas fluxes. We made the unexpected discovery that the size distributions for initially monodisperse foams do no monotonically broaden en route to steady state. Rather, the distribution bifurcates and exhibits a intermediary regime of bidispersity, where small bubbles decorate the vertices between four neighboring large bubbles. Only once the small bubbles fully evaporate is the steady state reached. This discovery was published during the current period, and our efforts focused on analyzing the data in the context of the drainage and the coarsening equations. The first step was to examine the vertical bubble speeds in steady state and to compare with the gas flux. We found that for the three highest fluxes that the two quantities were equal, meaning that the bubbles move upwards as a plug and that this motion accounts for all gas transport in the steady state. For the lowest gas flux, by contrast, the bubble speed is less than the gas flux, and decreases with height. This implies that gas transport is accomplished by a vertical diffusive flux between neighboring bubbles, which grows with height. The next step was to compare with the drainage equation. We found that the capillary term vanishes away from the bottom, once the bubble size distribution is polydisperse. Hence the downward pull of gravity on the fluid is balanced by the upward drag of the moving bubbles. The equation of force balance is consistent with a dissipation mechanism of Poisseuille flow in the Plateau borders with a no-slip boundary condition. It is also consistent with previous reports of both dimensionless permeability coefficient and of surface viscosity. The last step was to compare vertical bubble size profile with previous proposals for the liquid-fraction dependence of the coarsening rate. One novel feature is that in steady state we may compute d/dt by vertical velocity times d/dz. Another novel feature is that we can access foams that are ten times drier than possible with free drainage experiments. Our results rule out proposals in which gas flux across films is blocked by Plateau borders, but are consistent with a growth rate set by the characteristic pressure difference between bubbles and Plateau borders. Thus, overall, we bring new insight into a complex situation important in industry (eg froth flotation) which had not previously been amenable to straightforward analysis.

Task Progress:	(3) Speckle-Visibility Spectroscopy. In this thrust we focus on microscopic bubble-scale motion that fundamentally underlies the unusual flow properties of foams. At low shear rates, small clusters of bubbles suddenly rearrange from one packing configuration to another in discrete, separated, avalanche-like events. This is inferred from Diffusing-Wave Spectroscopy (DWS), which gives the event rate in terms of the average wait-time between events at a given location. But since DWS operates with a time-averaged signal, it cannot tell the distribution of wait times or the duration of the events or the bubble speeds during rearrangement. To access these features we have created a new multispeckle dynamic light scattering technique we call SVS. We published a long account of our method in Reviews of Scientific Instruments during the previous year. The basic idea is that the visibility of the speckle depends on the rate of motion within the sample in comparison to the duration of an exposure. We use this to extract the event duration and time between events, and accumulate the statistical distribution of these quantities. Furthermore, from the blurriness of the speckle we can extract the bubble speeds using our light scattering theory for SVS. None of these crucial quantities are available from DWS or from imaging of surface bubbles. Analysis of bubble speeds, event durations, the time lag between events, and the distributions thereof, is now almost complete. Currently we are comparing and contrasting with a “time resolved correlation” (TRC) analysis of the same raw data, and are preparing to write a draft. ----addendum for summer months: During this period my group consisted of Alex Gittings, Matthew Berck, and Dr. Klebert Feitosa. Alex is a graduate student, who will graduate soon; Klebert is a postdoc, whose term expired September 2007; Matthew is an undergraduate physics major. A new graduate student, Adam Roth, has just joined the project. And a new postdoc, Chris Jones from Noel Clark’s group at UC Boulder – Physics, will join in October. Over the summer and Fall, Alex Klebert and I made significant progress on several different aspects of the overall FOAM project: (1) I attended a meeting at ESA headquarters in June, where we redefined the scope of the FOAM mission in order to align with cost and engineering limitations. At my urging the mission now will focus on the physics of coarsening, and the dependence of the coarsening rate on the liquid fraction. Rheology aspects will be postponed for future flights. In consequence I helped formulate a new ESR (Experimental Requirements Document). The experiment features optical probes, like DTS, DWS, and now SVS as crucial diagnostics of the structure and dynamics. (2) I gave an invited presentation on bubble rearrangements and SVS at the annual colloid and interface science meeting of the American Chemical Society. (3) Steady-State Foams. As described in previous reports and publications, we produce foams in steady state by continuous bubbling of gas into a pool of surfactant solution at the bottom of a tall column. Our interest has been in time evolution via both drainage and coarsening, here combined in a novel previously unstudied way that also happens to be relevant to industrial processes like froth flotation for chemical and mechanical separations. This aspect of study, which was led by Klebert, is coming to a close, with a final publication newly submitted and under review. Besides this new work on dissemination of results, we have begun a new project with Matthew to investigate the mechanics of the foam. In particular we now drop a plastic bead into the steady foam, with size comparable to the bubbles, and image its behavior vs time. The key observation is that the bead height fluctuations around a steady-state average value. This height, and the know density difference and gas flux speed, give the mobility of the particle – ie a measure of the foam mechanics. Furthermore, fluctuations around the average are related to bubble-scale dynamical rearrangements, and potentially may yield frequency-dependent rheological properties of the foam. Ultimately this new knowledge will be key for helping define the next set of flight experiments on rheology vs liquid fraction. (4) Speckle-Visibility Spectroscopy. As described in previous reports, this focuses on microscopic bubble-scale motion that fundamentally underlies the unusual flow properties of foams. The method is essentially a time-resolved version of DWS, capable of following dynamics that change or evolve with time. By contrast, DWS relies on temporal averaging. Whereas DWS can reveal bubble rearrangement rates, SVS can also reveal bubble speeds and the duration of motion during individual rearrangement events. During the past four months, Alex has continued to analyze a massive data set on an evolving foam, with particular emphasis on understanding the baseline of the signal. This is more or less the final step. We have also prepared a set of figures and a rough outline for a publication. Alex will graduate when this paper is complete, hopefully before the end of 2007. (5) Tomography. Klebert and Adam together have begun a new project where we hope to obtain high-quality three-dimensional images of dry foams. Such data will form the basis for understanding structure at high liquid fractions, to be obtained in microgravity. Surprisingly, the best such images still date from 1945 on monodisperse foams. Our interest, of course, is to obtain even better data for polydisperse foam which have evolved by coarsening into a stationary state where the size distribution has a fixed shape and is independent of initial conditions. Toward this end, we have purchased a high-quality lens, a rotation stage, and lighting. We have assembled, aligned, and optimized the setup. We have created analysis programs. However these are slow and need to be improved. (6) Immediate plans. During the coming year, the group will consist of Matthew Berck (undergraduate), Adam Roth (graduate), and Chris Jones (postdoc). Matthew will continue the ball-in-steady-foam experiment. Adam will continue the tomography experiment. Chris will assist with both, and will begin more SVS experiments.
Bibliography Type:	Description: (Last Updated: 06/12/2025)
Abstracts for Journals and Proceedings	Feitosa F, Durian DJ. "Coarsening in a steady state foam." 2007 March Meeting of the American Physical Society. Bull. Am. Phys. Soc. 2007 Mar;MAR07, V29-1. , Mar-2007
Articles in Peer-reviewed Journals	Feitosa F, Durian DJ. "Gas and liquid transport in steady-state aqueous foam." European Journal of Physics E (to appear). , Nov-2007
Articles in Peer-reviewed Journals	Feitosa K, Halt OL, Kamien RD, Durian DJ "Bubble kinetics in a steady-state column of aqueous foam." Europhysics Letters 2006;76(4): 683-9. http://dx.doi.org/10.1209/epl/i2006-10304-5 , Nov-2006